

ABSHealthcareLite

Manual Quality Control and UAT Guide

MOD-01 — Company / Tenant Management

Version 1.0

Prepared: 19 July 2026

Environment: QC Docker (non-production)

Confidential — ABSHealthcareLite - QC Use Only

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MOD-01 - Company / Tenant Management

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****Credential handling:**** All passwords are ****Provided separately through secure channel****. Do not record passwords in this guide, screenshots, defect reports, chat, or test evidence.

1. Purpose, scope, and reference baseline

This guide defines repeatable manual QC and UAT for MOD-01 Company / Tenant Management. It verifies host administration, tenant isolation, tenant lifecycle, onboarding, branches, subscriptions, settings, auditability, security, database integrity, and user experience.

1.1 In scope

- Host authentication at `/host/login`` and tenant authentication at `/login``.
- Tenant CRUD, search, filtering, paging, status transitions, and onboarding.
- Tenant administration of branches, subscriptions, modules, usage limits, and settings.
- Audit trail, authorization boundaries, API validation, and PostgreSQL read-only checks.
- Responsive usability and observation-level performance checks.

1.2 Out of scope

- Production deployment, real payment collection, disaster recovery, and load/stress testing.
- Clinical workflows not reached through the MOD-01 tenant-management surface.
- Any destructive database operation, schema migration, or write SQL by QC.

1.3 Required reference documents

1. AI QC foundation report: ``docs/AI-QC/reports/001-Foundation.md``
2. Re-QC report: ``001-Foundation-ReQC-01.md`` — **AI Re-QC PASS 24/30**
3. QC environment guide: ``docs/QC/ABSHealthcareLite-Pilot-Docker-QC-Guide.md``

****Bangla explanation (বাংলা):**** Re-QC PASS 24/30 is a useful baseline, not a replacement for this manual test cycle. Every manual case in this guide must be executed and evidenced independently.

2. QC environment and access

Item	QC value
---	---
Application URL	<code>`http://192.168.2.44:3000`</code>
Reusable URL pattern	<code>`http://<QC_HOST>:3000`</code>
Health endpoint	<code>`GET /api/health`</code>
Host login route	<code>`/host/login`</code>
Tenant login route	<code>`/login`</code>
Host administrator	<code>`admin.abs`</code>
Tenant administrator	<code>`laila.hasan`</code>
App container	<code>`abs-healthcare-pilot-app`</code>
Database container	<code>`abs-healthcare-pilot-db`</code>
Compose database service	<code>`postgres`</code> (<code>`postgres:16-alpine`</code>)

Compose application service	`app` (from `Dockerfile`)
Database	`abs_healthcare_pilot`
PostgreSQL user	`abshealthcare`
Credentials	Provided separately through secure channel

2.1 Pre-test environment check

- 1. Open `http://<QC\_HOST>:3000/api/health` and capture the response.
- 2. Confirm that the app can reach PostgreSQL and no maintenance/error page is displayed.
- 3. Confirm the assigned browser uses a clean profile or a documented isolated profile.
- 4. Confirm the test date/time and tester identity in the execution log.
- 5. Confirm that only QC tenants are changed.

****Bangla explanation (বাংলা):**** Test শুরু করার আগে health endpoint এবং database connectivity যাচাই করুন। Shared QC environment-এ অন্য tester-এর data পরিবর্তন করবেন না; নিজস্ব test code ব্যবহার করুন।

3. Test data, roles, and execution rules

3.1 Naming convention

Data item	Convention / example
---	---
Tenant code	`QC01-YYYYMMDD-HHMM`
Tenant name	`MOD01 QC Tenant`
Branch code	`QC-BR-01`
Unique suffix	Current execution timestamp, for example `20260719-1958`
Host user	`admin.abs`
Tenant administrator	`laila.hasan`

Use a new timestamp suffix for each run. Record every generated tenant ID and code in the test execution log.

3.2 Roles

Role	Purpose
---	---
Host Admin	Creates, edits, activates, suspends, and reviews tenants
Tenant Admin	Manages the assigned tenant only
Unauthorized / Anonymous	Validates login and access rejection
Database Observer	Runs approved read-only SQL only

3.3 Execution rules

- 1. Execute cases in ID order unless a stated dependency requires otherwise.
- 2. Do not reuse an earlier test's authenticated browser session for isolation tests.
- 3. Set **Actual Result**, **Status**, **Evidence**, and **Tester Remarks** during execution.
- 4. Use `PASS`, `FAIL`, `BLOCKED`, `NOT APPLICABLE`, or `NOT RUN`; initial status is `NOT RUN`.
- 5. Stop a destructive or privacy-risking action and raise a defect if it reaches real data.

****Bangla explanation (বাংলা):**** প্রতিটি test case-এ Actual Result এবং evidence পূরণ বাধ্যতামূলক। FAIL হলে পরের dependent test চালানোর আগে defect ID লিখুন।

4. Test case record convention

Every row below is an individual test case. The following fields apply to **each row**:

Field	Completion instruction
---	---
Test Case ID	Immutable identifier with prefix `MOD01-`
Title	Case name
Objective	Verification goal

Role	Logged-in role or Anonymous
Preconditions	State required before execution
Test Data	Exact data or convention to use
Steps	Ordered action summary; expand in evidence notes if necessary
Expected Result	Pass condition
Actual Result	_____
Status	`NOT RUN`
Severity if failed	`Critical`, `High`, `Medium`, or `Low`
Screenshot/Evidence reference	_____
Tester remarks	_____

For compact PDF layout, tables use these abbreviations: **Pre** = Preconditions, **Data** = Test Data, **Exp** = Expected Result, **Act** = Actual Result, **Ev** = Screenshot/Evidence reference, **Remarks** = Tester remarks.

5. Authentication test cases

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---
MOD01-AUTH-001	Host login accepts valid credentials	Anonymous	Host account active	`admin.abs`; password separately	Open `/host/login`; submit valid credentials	Host dashboard opens; authenticated session created		NOT RUN	Critical		
MOD01-AUTH-002	Host login rejects invalid password	Anonymous	Login page open	`admin.abs`; invalid password	Submit invalid password	No login; safe generic error; no password disclosure		NOT RUN	High		
MOD01-AUTH-003	Host login rejects unknown username	Anonymous	Login page open	Unknown username	Submit credentials	No login; generic error; audit/security behavior as designed		NOT RUN	High		
MOD01-AUTH-004	Host login requires username	Anonymous	Login page open	Blank username	Submit without username	Client/server validation shown; request not authenticated		NOT RUN	Medium		
MOD01-AUTH-005	Host login requires password	Anonymous	Login page open	Blank password	Submit without password	Validation shown; password field remains masked		NOT RUN	Medium		
MOD01-AUTH-006	Tenant login accepts tenant admin	Anonymous	Tenant account active	`laila.hasan`; password separately	Open `/login`; submit valid credentials	Tenant landing page opens for assigned tenant		NOT RUN	Critical		
MOD01-AUTH-007	Tenant login rejects host credentials	Anonymous	Tenant login open	Host credentials	Submit host account at `/login`	Access rejected or safely routed per design; no tenant context		NOT RUN	High		
MOD01-AUTH-008	Host login rejects tenant credentials	Anonymous	Host login open	Tenant credentials	Submit tenant account at `/host/login`	Access rejected; no host dashboard		NOT RUN	High		
MOD01-AUTH-009	Password input is masked	Anonymous	Either login open	Any value	Type into password field	Characters are obscured; no value shown in UI		NOT RUN	Medium		

MOD01-AUTH-010	Logout terminates session	Host Admin	Host logged in	None	Logout; use browser Back; revisit protected route	Protected route requires login; cached sensitive content unavailable		NOT RUN	High		
MOD01-AUTH-011	Session survives normal navigation	Tenant Admin	Tenant logged in	None	Navigate between permitted tenant pages	Session remains valid; correct tenant context remains visible		NOT RUN	Medium		
MOD01-AUTH-012	Direct protected route redirects anonymous user	Anonymous	Clean browser session	Protected host and tenant URLs	Paste protected URLs	Redirect to correct login; intended target handled safely		NOT RUN	High		

6. Session isolation test cases

****Bangla explanation (বাংলা):**** Tenant isolation হলো MOD-01-এর প্রধান নিরাপত্তা শর্ত। Tenant A-এর URL, ID, cookie বা API request বদলে Tenant B-এর data দেখা বা পরিবর্তন করা যাবে না।

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---
MOD01-ISO-001	Tenant Admin sees only own tenant	Tenant Admin	Two QC tenants exist	Tenant A and B	Login as A; view tenant pages	Only Tenant A name, branches, modules, and settings appear		NOT RUN	Critical		
MOD01-ISO-002	Tenant URL ID tampering is denied	Tenant Admin	Tenant A logged in	Tenant B ID	Replace route/query ID with B ID	Denied/redir ected; no B data exposed		NOT RUN	Critical		
MOD01-ISO-003	Tenant API ID tampering is denied	Tenant Admin	Browser DevTools permitted	Tenant B ID	Replay permitted request with B identifier	403/404 or equivalent; no B response		NOT RUN	Critical		
MOD01-ISO-004	Host session cannot be mistaken for tenant session	Host Admin	Host logged in	Tenant URL	Visit tenant-only route	Correct authorization behavior; no accidental tenant impersonation		NOT RUN	High		
MOD01-ISO-005	Parallel tenant browser sessions remain separate	Tenant Admin	Separate browser profiles	Tenant A/B credentials	Login A and B in separate profiles; navigate	Each profile retains only its own context		NOT RUN	Critical		
MOD01-ISO-006	Logout clears tenant context	Tenant Admin	Tenant A logged in	None	Logout; login as B in same browser	No A data, menu state, or cached form data leaks to B		NOT RUN	High		
MOD01-ISO-007	Cross-tenant branch access is denied	Tenant Admin	Branch exists under B	B branch ID	Attempt direct B branch URL/API as A	Access denied; no update possible		NOT RUN	Critical		

7. Tenant list test cases

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
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---	---	---	---	---	---	---	---	---	---	---	---
MOD01-LIST-001	Open tenant list	Host Admin	Logged in	None	Open Company/Tenant list	List loads without error and shows permitted tenants		NOT RUN	High		
MOD01-LIST-002	List shows required columns	Host Admin	List loaded	None	Inspect headers and a row	Name/code/status and designated management fields are legible		NOT RUN	Medium		
MOD01-LIST-003	Search by exact tenant code	Host Admin	QC tenant exists	'QC01-YYYYMMDD-HHMM'	Search exact code	Matching tenant returned; unrelated rows excluded		NOT RUN	Medium		
MOD01-LIST-004	Search by partial tenant name	Host Admin	QC tenant exists	'MOD01 QC'	Search partial name	Expected matching rows returned		NOT RUN	Medium		
MOD01-LIST-005	Search handles no result	Host Admin	List loaded	Unique absent text	Search absent text	Clear empty-state message; no error		NOT RUN	Low		
MOD01-LIST-006	Status filter works	Host Admin	Active and suspended examples exist	Status value	Filter by status	Only selected status is shown		NOT RUN	Medium		
MOD01-LIST-007	Paging preserves valid result set	Host Admin	Enough records for paging	Page size/default	Change page or next page	Correct records/page state; no duplicate/omitted visible rows		NOT RUN	Medium		
MOD01-LIST-008	Open tenant details from list	Host Admin	QC tenant listed	QC tenant	Select row/action	Correct tenant detail page opens		NOT RUN	High		

8. Tenant create test cases

****Bangla explanation (বাংলা):**** Create test চলাকালে tenant code অবশ্যই unique হতে হবে। একই code দিয়ে second create করলে duplicate record বা partial onboarding তৈরি হওয়া গ্রহণযোগ্য নয়।

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---
MOD01-CREATE-001	Open create tenant form	Host Admin	Logged in	None	Choose Create Tenant	Form opens with clear required markers		NOT RUN	Medium		
MOD01-CREATE-002	Create valid minimal tenant	Host Admin	Form open	Unique code; name	Complete mandatory fields; save	One tenant created; success confirmation shown		NOT RUN	Critical		
MOD01-CREATE-003	Create valid full tenant	Host Admin	Form open	Name, code, contact, package	Complete all supported fields; save	Values persist accurately in details/list		NOT RUN	High		
MOD01-CREATE-004	Tenant code is required	Host Admin	Form open	Blank code	Complete other fields; save	Validation prevents save		NOT RUN	High		
MOD01-CREATE-005	Tenant name is required	Host Admin	Form open	Blank name	Complete other fields; save	Validation prevents save		NOT RUN	High		
MOD01-CREATE-006	Duplicate tenant code rejected	Host Admin	Existing QC code	Existing code	Create second tenant with same code	Clear duplicate validation; no duplicate		NOT RUN	Critical		

						row/DB record					
MOD01-CREATE-007	Leading/trailing spaces handled	Host Admin	Form open	Spaced name/code	Save with spaces around values	Safe trim or explicit validation; no ambiguous duplicate		NOT RUN	Medium		
MOD01-CREATE-008	Invalid code characters handled	Host Admin	Form open	Unsupported symbols	Save invalid code	Validation message; no malformed tenant created		NOT RUN	Medium		
MOD01-CREATE-009	Maximum field length handled	Host Admin	Form open	Boundary-length name/code	Submit at and above limit	Boundary accepted if valid; overflow rejected safely		NOT RUN	Medium		
MOD01-CREATE-010	Cancel does not create tenant	Host Admin	Populated unsaved form	Unique code	Click Cancel/Back; confirm leave if prompted	No tenant or onboarding artifacts created		NOT RUN	Medium		
MOD01-CREATE-011	Double submit is idempotent	Host Admin	Form valid	Unique code	Rapidly click Save / submit twice	One tenant created; UI prevents duplicate action		NOT RUN	Critical		
MOD01-CREATE-012	Create produces audit entry	Host Admin	Valid creation completed	New tenant code	Review audit history	Creation actor/time/entity/action recorded		NOT RUN	High		

9. Tenant edit test cases

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---
MOD01-EDIT-001	Edit tenant display name	Host Admin	Existing active QC tenant	Revised name	Edit and save	New name persists in detail and list		NOT RUN	High		
MOD01-EDIT-002	Edit permitted contact details	Host Admin	Existing tenant	Valid contact data	Edit permitted fields; save	Accurate persistence and display		NOT RUN	Medium		
MOD01-EDIT-003	Required field cannot be cleared	Host Admin	Edit form open	Blank required value	Clear and save	Validation prevents invalid state		NOT RUN	High		
MOD01-EDIT-004	Duplicate code edit rejected	Host Admin	Two QC tenants	Other tenant code	Change code to existing code	Rejected; original data remains intact		NOT RUN	Critical		
MOD01-EDIT-005	Cancel edit preserves saved values	Host Admin	Existing tenant	Unsaved change	Change; cancel/back	Persisted values remain unchanged		NOT RUN	Medium		
MOD01-EDIT-006	Edit produces audit and status history appropriately	Host Admin	Existing tenant	Revised value	Save edit; inspect history	Audit captures change; status history only changes when status changes		NOT RUN	High		

10. Onboarding test cases

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---

MOD01-ONB-001	Start onboarding for new tenant	Host Admin	Newly created tenant	QC tenant	Open onboarding action	Guided onboarding state opens for correct tenant		NOT RUN	High		
MOD01-ONB-002	Create initial branch	Host Admin	Onboarding active	'QC-BR-01'	Enter initial branch details	Branch is created and tied to tenant		NOT RUN	Critical		
MOD01-ONB-003	Select subscription package	Host Admin	Package available	Valid package	Select package; continue	Package selection persists		NOT RUN	High		
MOD01-ONB-004	Set tenant modules	Host Admin	Onboarding active	Enabled module set	Select approved modules	Module settings persist for tenant only		NOT RUN	High		
MOD01-ONB-005	Set usage limits	Host Admin	Onboarding active	Valid limits	Enter limits; continue	Limits validate and persist		NOT RUN	High		
MOD01-ONB-006	Mandatory onboarding step enforced	Host Admin	Step incomplete	Omit required value	Continue/finish	Cannot complete until requirement satisfied		NOT RUN	High		
MOD01-ONB-007	Invalid branch data is rejected	Host Admin	Branch step open	Invalid/blank branch code	Continue	Validation prevents invalid branch creation		NOT RUN	Medium		
MOD01-ONB-008	Back navigation retains entered values	Host Admin	Multiple steps entered	Valid draft values	Go Back then Next	Values retained unless intentionally changed		NOT RUN	Medium		
MOD01-ONB-009	Refresh/resume onboarding is safe	Host Admin	Incomplete onboarding	Existing draft	Refresh or reopen tenant	State is recoverable; no duplicate artifacts		NOT RUN	High		
MOD01-ONB-010	Complete onboarding activates usable setup	Host Admin	All valid steps complete	QC tenant	Finish onboarding; login/inspect	Expected tenant, branch, package, modules, limits available		NOT RUN	Critical		

11. Status lifecycle test cases

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---
MOD01-STATUS-001	View initial tenant status	Host Admin	New tenant exists	QC tenant	Open details	Status displayed consistently with lifecycle state		NOT RUN	Medium		
MOD01-STATUS-002	Activate eligible tenant	Host Admin	Valid configured tenant	QC tenant	Select Activate; confirm	Status changes to Active and is persisted		NOT RUN	Critical		
MOD01-STATUS-003	Suspend active tenant	Host Admin	Active QC tenant	Reason if required	Select Suspend; confirm	Status becomes Suspended; access follows policy		NOT RUN	Critical		
MOD01-STATUS-004	Suspended tenant login follows policy	Tenant Admin	Tenant suspended	Tenant credentials	Attempt '/login'	Login/access blocked or restricted per approved policy		NOT RUN	Critical		

MOD01-STATUS-005	Reactivate suspended tenant	Host Admin	Suspended tenant	QC tenant	Select Activate/reactivate	Tenant returns to correct usable state		NOT RUN	Critical		
MOD01-STATUS-006	Invalid lifecycle transition denied	Host Admin	Tenant in state with invalid target	Target status	Attempt invalid transition	Action unavailable or denied without data damage		NOT RUN	High		
MOD01-STATUS-007	Status change creates history	Host Admin	Transition completed	QC tenant	Inspect status history	From/to status, actor, timestamp, and reason captured if supported		NOT RUN	High		
MOD01-STATUS-008	Status change creates audit event	Host Admin	Transition completed	QC tenant	Inspect audit log	Traceable status action recorded		NOT RUN	High		

12. Branch test cases

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---
MOD01-BRANCH-001	View branch list	Tenant Admin	Tenant logged in	None	Open Branches	Own tenant branches load correctly		NOT RUN	High		
MOD01-BRANCH-002	Create valid branch	Tenant Admin	Branch permission available	`QC-BR-01`	Create branch with valid fields	Branch created under current tenant		NOT RUN	High		
MOD01-BRANCH-003	Branch code required	Tenant Admin	Create form open	Blank code	Save	Validation prevents save		NOT RUN	Medium		
MOD01-BRANCH-004	Branch name required	Tenant Admin	Create form open	Blank name	Save	Validation prevents save		NOT RUN	Medium		
MOD01-BRANCH-005	Duplicate branch code rejected	Tenant Admin	Existing branch	Existing `QC-BR-01`	Create same code	Duplicate rejected within tenant scope		NOT RUN	High		
MOD01-BRANCH-006	Edit branch details	Tenant Admin	Existing branch	Revised name	Edit and save	Revised permitted fields persist		NOT RUN	Medium		
MOD01-BRANCH-007	Cancel branch edit	Tenant Admin	Existing branch	Unsaved change	Edit; cancel	No unsaved change persists		NOT RUN	Low		
MOD01-BRANCH-008	Deactivate/active branch lifecycle	Tenant Admin	Existing active branch	Branch	Change branch status if supported	State persists and UI/access follows policy		NOT RUN	High		
MOD01-BRANCH-009	Branch cannot cross tenant boundary	Tenant Admin	Tenant A/B branches exist	B branch ID	Tamper direct URL/API	B branch not viewed or changed		NOT RUN	Critical		
MOD01-BRANCH-010	Branch action writes audit trail	Tenant Admin	Branch change complete	Branch	Review audit data	Action is attributable and tenant-scoped		NOT RUN	High		

13. Subscription test cases

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---

MOD01-SUB-001	View available subscription packages	Host Admin	Packages seeded	None	Open package/subscription view	Available packages are listed accurately		NOT RUN	Medium		
MOD01-SUB-002	Assign valid package	Host Admin	Active tenant; package exists	Valid package	Assign package; save	Tenant subscription created/updated		NOT RUN	Critical		
MOD01-SUB-003	Subscription dates validate	Host Admin	Assignment form open	End before start	Save invalid dates	Validation prevents invalid period		NOT RUN	High		
MOD01-SUB-004	Required package selection enforced	Host Admin	Assignment form open	No package	Save	Validation shown; no partial subscription		NOT RUN	High		
MOD01-SUB-005	Change package records effective result	Host Admin	Existing subscription	Different package	Change package	Correct current/effective subscription displayed		NOT RUN	High		
MOD01-SUB-006	Enabled modules follow configuration	Host Admin	Subscription assigned	Module set	View/manage modules	Tenant module records reflect intended selection		NOT RUN	High		
MOD01-SUB-007	Usage limits accept valid boundary	Host Admin	Limits view open	Valid min/max	Save limits	Values persist and are displayed consistently		NOT RUN	High		
MOD01-SUB-008	Negative/invalid usage limit rejected	Host Admin	Limits view open	Negative/text value	Save	Validation prevents invalid numeric value		NOT RUN	High		
MOD01-SUB-009	Expired subscription behavior visible	Host Admin	Controlled expired QC subscription	Expired date	Open tenant/access view	Expiry status and restrictions follow policy		NOT RUN	Critical		
MOD01-SUB-010	Subscription change audited	Host Admin	Change completed	QC tenant	Inspect audit log	Package/modules/limits action traceable		NOT RUN	High		

14. Settings test cases

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---
MOD01-SET-001	Open tenant settings	Tenant Admin	Tenant logged in	None	Open Settings	Correct tenant settings load		NOT RUN	High		
MOD01-SET-002	Save valid tenant setting	Tenant Admin	Settings open	Valid value	Change supported setting; save	Value persists after refresh		NOT RUN	High		
MOD01-SET-003	Required setting validation	Tenant Admin	Settings open	Blank required value	Save	Validation shown; old value retained		NOT RUN	Medium		
MOD01-SET-004	Invalid format validation	Tenant Admin	Settings open	Invalid supported format	Save	Field-level safe validation shown		NOT RUN	Medium		
MOD01-SET-005	Cancel preserves setting	Tenant Admin	Settings open	Unsaved value	Change; cancel	Saved setting remains unchanged		NOT RUN	Low		

MOD01-SET-006	Settings are tenant isolated	Tenant Admin	Tenant A/B exist	Tenant A setting	Change A; inspect B	B setting unaffected		NOT RUN	Critical		
MOD01-SET-007	Unauthorized setting route denied	Anonymous	Clean session	Settings URL	Paste protected settings URL	Login/authentication barrier applied		NOT RUN	High		
MOD01-SET-008	Setting change audited	Tenant Admin	Setting saved	Changed value	Inspect audit trail	Actor, time, target, and action recorded		NOT RUN	High		

15. Audit test cases

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---
MOD01-AUDIT-001	Audit list opens for authorized role	Host Admin	Logged in	None	Open audit view	Audit records load without exposure beyond role		NOT RUN	High		
MOD01-AUDIT-002	Tenant creation audit exists	Host Admin	Create performed	New tenant	Filter/find create event	Actor/action/entity/time captured		NOT RUN	High		
MOD01-AUDIT-003	Tenant edit audit exists	Host Admin	Edit performed	QC tenant	Find edit event	Meaningful change event captured		NOT RUN	High		
MOD01-AUDIT-004	Status audit exists	Host Admin	Status change performed	QC tenant	Find status event	Lifecycle event recorded		NOT RUN	High		
MOD01-AUDIT-005	Branch creation audit exists	Tenant Admin	Branch create performed	`QC-BR-01`	Find event	Tenant-scoped branch action captured		NOT RUN	High		
MOD01-AUDIT-006	Subscription audit exists	Host Admin	Subscription change performed	QC tenant	Find event	Subscription action captured		NOT RUN	High		
MOD01-AUDIT-007	Settings audit exists	Tenant Admin	Setting change performed	Setting key	Find event	Setting action captured without secret disclosure		NOT RUN	High		
MOD01-AUDIT-008	Audit timestamp is plausible	Host Admin	Events available	Current time	Compare event timestamp	Timestamp/timezone follows configured standard		NOT RUN	Medium		
MOD01-AUDIT-009	Audit actor is correct	Host Admin	Event performed by known user	User identity	Compare actor	Actor matches authenticated initiator		NOT RUN	High		
MOD01-AUDIT-010	Audit filtering/search works	Host Admin	Multiple events	Entity/action/text	Apply filters/search	Intended records returned without unrelated leak		NOT RUN	Medium		
MOD01-AUDIT-011	Tenant user cannot view other tenant audit	Tenant Admin	Tenant A/B data exists	B identifier	Attempt list/URL/API access	No B audit details exposed		NOT RUN	Critical		
MOD01-AUDIT-012	Audit is not editable through UI	Host Admin	Audit record visible	Existing record	Look for edit/delete; attempt permitted UI paths	No unauthorized alteration/deletion capability		NOT RUN	Critical		

16. Security test cases

****Bangla explanation (বাংলা):**** এই security tests-এ ক্ষতিকর payload ব্যবহার করবেন না। শুধুমাত্র controlled QC account ও harmless validation strings ব্যবহার করুন। Sensitive response, token, password, বা personal data evidence-এ redact করুন।

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---
MOD01-SEC-001	Anonymous host route protection	Anonymous	Clean session	Host route	Open protected host URL	Redirect/deny; no sensitive content		NOT RUN	Critical		
MOD01-SEC-002	Anonymous tenant route protection	Anonymous	Clean session	Tenant route	Open protected tenant URL	Redirect/deny; no tenant content		NOT RUN	Critical		
MOD01-SEC-003	Host-only action denied to tenant user	Tenant Admin	Tenant logged in	Host action URL	Attempt direct navigation	Access denied; no action executed		NOT RUN	Critical		
MOD01-SEC-004	Tenant-only boundary enforced	Tenant Admin	A/B tenants exist	B tenant/branch ID	Tamper route identifier	No cross-tenant read/write		NOT RUN	Critical		
MOD01-SEC-005	API rejects missing authentication	Anonymous	API endpoint known	Protected request	Send/replay without session	401/redirect/safe denial; no data		NOT RUN	Critical		
MOD01-SEC-006	API rejects insufficient authorization	Tenant Admin	A logged in	Host/B resource	Request protected resource	403/404 or safe equivalent		NOT RUN	Critical		
MOD01-SEC-007	Login error avoids account enumeration	Anonymous	Login open	Known vs unknown user	Compare invalid-login responses	Response is equivalently safe		NOT RUN	High		
MOD01-SEC-008	Password never appears in page/source evidence	Anonymous	Login page	Test credential	Inspect UI/network safely	Password masked; not rendered/logged in plain view		NOT RUN	Critical		
MOD01-SEC-009	Input validation handles HTML-like text	Host Admin	Safe disposable field	Harmless markup text	Enter/save/view field	Encoded/displayed safely; no script execution		NOT RUN	High		
MOD01-SEC-010	Input validation handles quote characters	Host Admin	Disposable field	Quotes/apostrophes	Enter/save/search	No server error, injection symptom, or corrupted query		NOT RUN	High		
MOD01-SEC-011	CSRF-sensitive change behavior checked	Host Admin	Authenticated session	State-changing form	Inspect normal request/submit from app	Change has expected anti-forgery/session safeguards		NOT RUN	High		
MOD01-SEC-012	Logout invalidates protected API use	Host Admin	Logged in then logged out	Previous route	Reuse/back/replay normal request	Authorization not retained after logout		NOT RUN	High		
MOD01-SEC-013	Error responses do not expose internals	Anonymous	Invalid input route	Invalid ID/payload	Trigger safe validation error	No stack trace, SQL, filesystem, secret, or token disclosed		NOT RUN	High		
MOD01-SEC-014	Security headers/cookie attributes observed	Anonymous	Browser DevTools permitted	Login response	Inspect response/cookie metadata	Secure configuration observed per environment		NOT RUN	Medium		

						capability; record gaps					
--	--	--	--	--	--	----------------------------	--	--	--	--	--

17. Database verification test cases

Only the Database Observer performs these tests. All queries are read-only. Never run `INSERT`, `UPDATE`, `DELETE`, `TRUNCATE`, DDL, or migration commands during QC.

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---
MOD01-DB-001	Tenant record created	Database Observer	Create completed	`:tenant_code`	Run Q1	One expected tenant row exists		NOT RUN	Critical		
MOD01-DB-002	Branch belongs to tenant	Database Observer	Branch created	`:tenant_id`	Run Q2	Branch rows carry correct tenant foreign key		NOT RUN	Critical		
MOD01-DB-003	Subscription record exists	Database Observer	Subscription assigned	`:tenant_id`	Run Q3	Expected subscription /package relationship exists		NOT RUN	High		
MOD01-DB-004	Tenant modules are scoped	Database Observer	Modules configured	`:tenant_id`	Run Q4	Module rows match selected configuration		NOT RUN	High		
MOD01-DB-005	Usage limits are persisted	Database Observer	Limits saved	`:tenant_id`	Run Q5	Stored limits match permitted UI values		NOT RUN	High		
MOD01-DB-006	User is tenant scoped	Database Observer	Tenant user exists	`:tenant_id`	Run Q6	User record is associated with expected tenant		NOT RUN	Critical		
MOD01-DB-007	Audit record exists	Database Observer	Change performed	`:tenant_id`	Run Q7	Audit action references expected tenant/event		NOT RUN	High		
MOD01-DB-008	Status history is chronological	Database Observer	Status changed	`:tenant_id`	Run Q8	Status transitions are present in chronological order		NOT RUN	High		
MOD01-DB-009	No unexpected duplicate tenant code	Database Observer	Create/duplicate test completed	`:tenant_code`	Run Q9	Count is exactly one for valid created code		NOT RUN	Critical		

17.1 Approved read-only PostgreSQL queries

Connect only to database `abs\_healthcare\_pilot` with user `abshealthcare`; password is **Provided separately through secure channel**.

```

-- Q1: tenant record and status
SELECT id, tenant_code, tenant_name, tenant_status, onboarding_status, created_at, updated_at
FROM tenants
WHERE tenant_code = :tenant_code;

-- Q2: branches scoped to the tested tenant
SELECT id, tenant_id, code, name, status, is_active, created_at
FROM branches
WHERE tenant_id = :tenant_id
ORDER BY created_at;

-- Q3: subscriptions and package reference
SELECT ts.id, ts.tenant_id, ts.package_id, ts.subscription_status,
       ts.subscription_start, ts.subscription_end, ts.billing_cycle,
       sp.package_code, sp.package_name
FROM tenant_subscriptions ts
LEFT JOIN subscription_packages sp ON sp.id = ts.package_id
WHERE ts.tenant_id = :tenant_id AND ts.is_active = true
ORDER BY ts.created_at DESC;

-- Q4: enabled modules for tenant (via module_registry join)
SELECT tm.id, tm.tenant_id, mr.module_code, mr.module_name,
       tm.is_enabled, tm.module_status, tm.created_at
FROM tenant_modules tm
JOIN module_registry mr ON mr.id = tm.module_id
WHERE tm.tenant_id = :tenant_id AND tm.is_active = true
ORDER BY mr.module_code;

-- Q5: tenant usage limits (1:1 row per tenant)
SELECT id, tenant_id, max_branches, max_users, max_orders_per_month,
       max_reports_per_month, max_storage_gb, created_at, updated_at
FROM tenant_usage_limits
WHERE tenant_id = :tenant_id;

-- Q6: users assigned to tested tenant (do not export password_hash)
SELECT id, tenant_id, username, email, user_status, is_host_admin, created_at
FROM users
WHERE tenant_id = :tenant_id AND is_host_admin = false
ORDER BY username;

-- Q7: tenant-related audit events
SELECT id, tenant_id, user_id, action_type, entity_type, entity_id, created_at, created_by
FROM audit_logs
WHERE tenant_id = :tenant_id
ORDER BY created_at DESC
LIMIT 50;

-- Q8: tenant status lifecycle history
SELECT id, tenant_id, entity_type, entity_id, old_status, new_status,
       remarks, changed_by, changed_at
FROM status_histories
WHERE tenant_id = :tenant_id AND entity_type = 'Tenant'
ORDER BY changed_at;

-- Q9: duplicate tenant code check
SELECT tenant_code, COUNT(*) AS tenant_count
FROM tenants
WHERE tenant_code = :tenant_code
GROUP BY tenant_code;

-- Q10: duplicate branch code within tenant
SELECT code, COUNT(*) AS branch_count
FROM branches
WHERE tenant_id = :tenant_id AND code = :branch_code
GROUP BY code;

-- Q11: orphan subscription check (should return zero rows)
SELECT ts.id, ts.tenant_id
FROM tenant_subscriptions ts
LEFT JOIN tenants t ON t.id = ts.tenant_id
WHERE t.id IS NULL;

-- Q12: latest audit entries (host platform or tenant)
SELECT id, tenant_id, action_type, entity_type, created_at, created_by
FROM audit_logs
ORDER BY created_at DESC
LIMIT 20;

```

****Bangla explanation (বাংলা):**** SQL placeholders `:tenant\_code` এবং `:tenant\_id` execution-এর আগে প্রকৃত QC value দিয়ে replace করুন। Query শুধু read-only; result-এ password hash, token, বা sensitive column থাকলে evidence-এ include করবেন না।

18. UI/UX test cases

Test Case ID	Title / Objective	Role	Pre	Data	Steps	Exp	Act	Status	Severity if failed	Ev	Remarks
---	---	---	---	---	---	---	---	---	---	---	---
MOD01-UI-001	Host login layout is usable	Anonymous	Desktop browser	None	Inspect `/host/login`	Labels, inputs, button, and error area are clear		NOT RUN	Low		
MOD01-UI-002	Tenant login layout is usable	Anonymous	Desktop browser	None	Inspect `/login`	Correct tenant login context is clear		NOT RUN	Low		
MOD01-UI-003	Tenant list headers align with data	Host Admin	List loaded	None	Inspect table	Headers/data/actions readable and aligned		NOT RUN	Low		
MOD01-UI-004	Empty-state messaging is actionable	Host Admin	Search no-result state	Absent search text	Search	User understands no results and recovery action		NOT RUN	Low		
MOD01-UI-005	Validation messages are understandable	Host Admin	Create form	Blank required field	Submit	Message identifies field and correction without jargon		NOT RUN	Low		
MOD01-UI-006	Required fields are visibly identified	Host Admin	Create/edit form	None	Inspect form	Required indicators are consistent and visible		NOT RUN	Low		
MOD01-UI-007	Save feedback is visible	Host Admin	Valid form	QC update	Save	Clear success/failure feedback prevents uncertainty		NOT RUN	Medium		
MOD01-UI-008	Destructive status action has confirmation	Host Admin	Active tenant	Suspend action	Initiate suspend	Confirmation explains impact and allows cancel		NOT RUN	Medium		
MOD01-UI-009	Keyboard tab order is logical	Host Admin	Form open	None	Tab through controls	Focus order follows visual/workflow order		NOT RUN	Low		
MOD01-UI-010	Visible focus indicator exists	Host Admin	Form open	None	Navigate with keyboard	Focused control is visually identifiable		NOT RUN	Medium		
MOD01-UI-011	Enter key behavior is safe	Host Admin	Populated form	Valid draft	Press Enter in fields	No unintended destructive/double action		NOT RUN	Medium		
MOD01-UI-012	Page refresh behavior is understandable	Host Admin	Form/list state	Draft/search	Refresh	State is safely retained, reset, or warned as designed		NOT RUN	Medium		
MOD01-UI-013	Mobile-width layout is usable	Host Admin	Narrow viewport	None	Inspect core pages at mobile width	No critical overlap/cutoff; controls usable		NOT RUN	Medium		

MOD01-UI-014	Tablet-width layout is usable	Host Admin	Medium viewport	None	Inspect core pages	Tables/actions remain usable		NOT RUN	Low		
MOD01-UI-015	Long tenant name is handled	Host Admin	Test tenant available	Long valid name	View list/detail	No clipping of critical content or broken layout		NOT RUN	Low		
MOD01-UI-016	Status uses clear visual meaning	Host Admin	Different statuses exist	Active/suspended	Inspect badges/text	Status is distinguishable without color alone where practical		NOT RUN	Low		
MOD01-UI-017	Error page is user-safe	Anonymous	Safe invalid route/input	Invalid route	Navigate safely	Friendly recovery guidance; no internals exposed		NOT RUN	Medium		
MOD01-UI-018	Browser back navigation is safe	Host Admin	Save action complete	None	Use Back after save	No duplicate submit or misleading stale state		NOT RUN	Medium		

19. Performance observation table

Performance observations are not load tests. Use one normal QC browser and record approximate elapsed time, network condition, and evidence.

Observation ID	Scenario	Target observation	Actual time	Status	Evidence / remarks
---	---	---	---	---	---
MOD01-PERF-001	`GET /api/health`	Responds successfully without visible delay		NOT RUN	
MOD01-PERF-002	Host login	Dashboard available in acceptable QC time		NOT RUN	
MOD01-PERF-003	Tenant login	Tenant landing page available in acceptable QC time		NOT RUN	
MOD01-PERF-004	Tenant list initial load	List/action controls become usable promptly		NOT RUN	
MOD01-PERF-005	Tenant search	Result/empty state updates promptly		NOT RUN	
MOD01-PERF-006	Create tenant	Save completes once without extended spinner/error		NOT RUN	
MOD01-PERF-007	Tenant detail/onboarding open	View opens without broken partial state		NOT RUN	
MOD01-PERF-008	Audit list/filter	Records/filter response remains usable		NOT RUN	

Record a defect only when a delay materially prevents normal QC use, causes timeout, causes duplicate submission, or produces inconsistent data. Include timestamp and browser/network observation.

20. Evidence handling and naming

20.1 Required evidence

- Capture the initial state, material action, and final result for every PASS or FAIL where practical.
- For security and privacy cases, redact usernames other than permitted test IDs and never capture passwords, session tokens, or secrets.
- Save evidence in the approved QC evidence location; reference the relative name in the case row.

20.2 Naming standard

Use:

MOD01-TC001-Step01.png
MOD01-TC001-Step02.png
MOD01-TC001-HealthResponse.txt
MOD01-TC042-QueryQ2.csv
MOD01-TC085-DefectABS-123.png

Where `TC001` is the sequential execution case number, not necessarily the suffix in the test-case ID. Use `.png` for screenshots, `.txt` for sanitized response text, and `.csv` only for approved non-sensitive query output.

****Bangla explanation (বাংলা):**** Screenshot file name একরকম রাখুন যাতে test case, step এবং defect সহজে trace করা যায়। Password, token, cookie, full connection string, বা sensitive user data দেখা গেলে screenshot redact করুন।

21. Defect template

Create one defect per independently reproducible issue. Do not combine unrelated failures.

Field	Required content
---	---
Defect ID	Tracker-generated ID
Title	Concise impact-focused title
Module	`MOD-01 Company / Tenant Management`
Test Case ID	For example `MOD01-SEC-004`
Environment	QC Docker, URL, build/commit if available
Severity	Critical / High / Medium / Low
Priority	Project triage value
Preconditions	Exact account/state/data setup
Steps to reproduce	Numbered, deterministic steps
Actual result	What occurred, including safe error text
Expected result	Requirement/pass condition
Impact	Tenant, user, data, security, or workflow impact
Evidence	Evidence filenames and sanitized logs
Reproducibility	Always / Intermittent / Once
Reporter and date	Tester identity and date/time
Retest result	Pass / Fail / Blocked with evidence

21.1 Severity guidance

Severity	Use when
---	---
Critical	Cross-tenant exposure/change, authentication bypass, data corruption, or service unavailable
High	Core lifecycle/create/subscription failure, authorization failure, or substantial audit failure
Medium	Validation, persistence, workflow, or UX issue with workable mitigation
Low	Cosmetic, wording, alignment, or minor usability issue

22. Entry and exit criteria

22.1 Entry criteria

- QC URL and `GET /api/health` are available.
- `abs-healthcare-pilot-app` and `abs-healthcare-pilot-db` are running.
- Approved host and tenant accounts are available through secure channel.
- At least two separable QC tenant contexts can be used for isolation tests.
- Subscription packages and necessary baseline data are available.
- Tester has approved read-only PostgreSQL access or a designated Database Observer is assigned.
- Evidence storage and defect tracker are available.

22.2 Exit criteria

- All applicable MOD01 test cases have an execution status and evidence reference or documented reason.
- All Critical failures are fixed and retested PASS, or formally accepted by authorized release authority.
- All High failures have an agreed disposition, owner, and retest/release decision.
- Tenant isolation, authentication, lifecycle, and audit high-risk cases pass.
- Database verification confirms the tested data relationships and no unexpected duplicate test tenant code remains.
- Test summary, defect list, and UAT sign-off are completed.

****Bangla explanation (বাংলা):**** Exit criteria পূরণ না হলে QC PASS ঘোষণা করা যাবে না। বিশেষ করে Critical security বা tenant isolation defect থাকলে release sign-off গ্রহণযোগ্য নয়।

23. Test summary and UAT sign-off

23.1 Execution summary

Metric	Count
---	---
Planned functional/security/database/UI cases	144
Performance observations	8
Passed	
Failed	
Blocked	
Not run	
Not applicable	
Open Critical defects	
Open High defects	

23.2 Sign-off decision

Decision	Select one
---	---
QC Pass	☐
QC Pass with accepted risks	☐
QC Fail / not ready for UAT	☐
UAT Accepted	☐
UAT Rejected / deferred	☐

23.3 Sign-off page

Sign-off role	Name	Signature / approval reference	Date	Comments
---	---	---	---	---
Independent QC Engineer				
QC Lead				
Development Representative				
Product / Business Owner				
UAT Representative				
Release Authority				

23.4 Final declaration

I confirm that the test execution recorded in this document was performed in the stated QC Docker non-production environment, using controlled QC data. Evidence and defects have been linked where required. No plaintext password has been recorded in this distributable document.